

Problem

Show that for any positive integers a and b , the number

$$(36a + b)(a + 36b)$$

cannot be a power of 2.

Source: **APMO 1998**

§1 Assuming a Power of Two Exists

Suppose, for contradiction, that

$$(36a + b)(a + 36b)$$

is a power of 2.

Since both factors are positive integers, each factor must itself be a power of 2. Thus there exist integers

$$m > n \geq 0$$

such that

$$36a + b = 2^m =: r, \quad a + 36b = 2^n =: s.$$

We therefore obtain the linear system

$$36a + b = r, \quad a + 36b = s.$$

Subtracting suitable multiples gives

$$36r - s = 1295a = 35 \cdot 37a,$$

and

$$36s - r = 1295b = 35 \cdot 37b.$$

Consequently,

$$\frac{1}{36} < \frac{s}{r} = 2^{n-m} < 36.$$

Since $r > s$, it follows that

$$0 < m - n < 6.$$

§2 A Congruence Modulo 37

Consider

$$r^2 - s^2 = (r - s)(r + s).$$

Using the expressions above,

$$r^2 - s^2 = (36r - s)b - (36s - r)a.$$

A more direct computation yields

$$r^2 - s^2 = 35 \cdot 37 (a^2 - b^2).$$

Since

$$r = 2^m, \quad s = 2^n,$$

we obtain

$$2^{2n}(2^{2(m-n)} - 1) = 35 \cdot 37 (a^2 - b^2).$$

Because 37 is odd, it follows that

$$37 \mid (2^{2(m-n)} - 1).$$

Hence

$$4^{m-n} \equiv 1 \pmod{37}.$$

§3 The Order of Four Modulo 37

We compute successive powers of 4 modulo 37:

$$4^1 \equiv 4, \quad 4^2 \equiv 16, \quad 4^3 \equiv 27, \quad 4^4 \equiv 34,$$

$$4^5 \equiv 25, \quad 4^6 \equiv 26, \quad 4^7 \equiv 30, \quad 4^8 \equiv 9,$$

$$4^9 \equiv 36 \equiv -1 \pmod{37}.$$

Therefore

$$4^{18} \equiv 1 \pmod{37},$$

and no smaller positive exponent yields 1.

Thus the order of 4 modulo 37 is

$$18.$$

Since

$$4^{m-n} \equiv 1 \pmod{37},$$

the order condition implies

$$18 \mid (m - n).$$

§4 The Contradiction

Earlier we established that

$$0 < m - n < 6.$$

However,

$$18 \mid (m - n)$$

forces

$$m - n \geq 18,$$

unless $m - n = 0$.

Neither possibility is compatible with

$$0 < m - n < 6.$$

This contradiction shows that our original assumption was impossible.

§5 The Final Conclusion

Therefore there do not exist positive integers a and b for which

$$(36a + b)(a + 36b)$$

is a power of 2.

Hence

$(36a + b)(a + 36b)$ is never a power of 2.